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ODD END TERM EXAMINATION DECEMBER - 2024

Name of the Course: **B.Tech.,**
Name of the Paper: **Automobile Engineering**

Semester: **7th**
Paper Code: **(TME 707)**

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20 marks)	
(a)	Discuss the role of resistances to vehicle motion in determining the design of a gearbox. (CO1)	CO1, CO2, and CO5
(b)	Explain the materials and functional requirements of critical engine components like cylinder heads and connecting rods. (CO2)	
(c)	Analyze the advantages of various chassis designs in improving vehicle stability. (CO5)	
Q2	(20 marks)	
(a)	Explain the working principles of the overdrive system and its impact on fuel efficiency. (CO1)	CO1, CO3, and CO4
(b)	Differentiate between gear shift mechanisms used in manual and automatic transmissions. (CO3)	
(c)	Design a conceptual model for integrating a transfer box in off-road vehicles. (CO4)	
Q3	(20 marks)	
(a)	Illustrate the role of electronically controlled diesel injection systems in reducing emissions. (CO1)	CO1, CO2, and CO3
(b)	Compare the advantages of three-way catalytic converters and turbochargers in emission control and engine performance. (CO2)	
(c)	Suggest improvements for an electronic ignition system to enhance fuel efficiency. (CO3)	
Q4	(20 marks)	
(a)	Describe the components and functionality of power steering systems. (CO1)	CO1, CO4, and CO6
(b)	Analyze the performance differences between pneumatic and hydraulic suspension systems. (CO4)	
(c)	Evaluate the effectiveness of traction control systems in modern braking mechanisms. (CO6)	
Q5	(20 marks)	
(a)	Explain the modifications required to use bio-diesel in CI engines. (CO1)	CO1, CO5, and CO6
(b)	Discuss the performance and combustion characteristics of gasohol in SI engines. (CO2)	
(c)	Propose an innovative design for a hybrid electric vehicle focusing on energy efficiency. (CO3)	

